

Lesson 3

AI & Creativity

Can Machines Be Artists?

Grade 4 · Academic Year 2025–26

Learning Objectives

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| 1 | Explain what Artificial Intelligence (AI) is in simple terms. |
| 2 | Describe how AI learns from examples to create new content. |
| 3 | Identify real-world examples of AI-generated art, music, and writing. |
| 4 | Compare what humans do best creatively versus what AI does best. |
| 5 | Reflect on the ethical question: Can machines truly be artists? |

What Is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that gives computers the ability to learn and make decisions — a little like how humans think. Instead of following rigid rules, AI programs learn from huge amounts of data (examples) and improve over time.

AI can already:

- Recognise faces and objects in photos.
- Understand spoken language and translate it.
- Compose original music and generate images.
- Write stories, poems, and even code.
- Beat world champions at chess and other games.

■ *Think of AI as a super-fast learner that has studied millions of examples and can now create something new from what it learned.*

What Is Creativity?

Creativity is the human ability to use imagination to make something new — a painting, a song, a story, a dance. Creativity is driven by feelings, experiences, and the desire to communicate something meaningful to others.

Human creativity in action:

■ Visual Art: Painting, drawing, sculpture, photography	■ Performing: Dance, theatre, stand-up comedy
■ Music: Composing, singing, playing instruments	■ Design: Architecture, fashion, product design
👉 ■ Writing: Stories, poems, scripts, songs	■ Innovation: Inventing new tools and solutions

How Does AI Make Art?

AI art tools work by studying millions of existing artworks and learning the patterns — colours, shapes, brushstrokes, styles. Then, when you give the AI a text description (a 'prompt'), it combines what it has learned to generate a brand-new image, song, or story.

Famous AI creative tools:

DALL·E / Midjourney	Type a description and AI generates a unique image in seconds. E.g. 'A purple elephant playing cricket at sunset'.
AIVA	An AI composer that creates original music for films, games, and advertisements after studying classical pieces.
ChatGPT	An AI that can write poems, stories, essays, and even code based on prompts you give it.
Next Rembrandt	An AI project that studied 346 Rembrandt paintings and generated a brand-new portrait in his exact style.

Types of AI Art

What AI Can Create

Type	What AI Does	Real Example
Paintings & Drawings	Generates images from text descriptions	DALL·E, Midjourney, Stable Diffusion
Music & Songs	Composes melodies and harmonies	AIVA, Suno AI
Stories & Poems	Writes creative text from prompts	ChatGPT, Claude
Video & Animation	Creates short clips from descriptions	Sora by OpenAI

Voice & Speech

Generates realistic human
voices

ElevenLabs

Humans vs AI — Who Does It Better?

A fair comparison:

WHAT HUMANS DO BEST	WHAT AI DOES BEST
Feel genuine emotions & express them	Process millions of examples instantly
Draw on personal memories & experiences	Mix ideas from many styles at once
Understand cultural context & meaning	Produce hundreds of versions rapidly
Choose to make art for a reason or cause	Work 24/7 without fatigue
Break rules intentionally for effect	Perfectly replicate any specific style

*"The artist brings imagination and feelings. AI helps with patterns and speed.
Together — they are an amazing team!"*

The Big Question — Can Machines Be Artists?

This is one of the most exciting debates in technology and philosophy today. There are thoughtful arguments on both sides.

YES — AI can be considered an artist:

- AI produces genuinely new works that did not exist before.
- Some AI art has sold for hundreds of thousands of dollars at auction.
- AI responds to prompts with creative variation, not just copying.

NO — AI is just a very clever tool:

- AI has no feelings, no lived experience, and no intention.
- AI cannot choose to make art — it responds to human prompts.
- The creativity comes from the human who designs the prompt and selects the output.

— ■ Activity: Class Discussion

Question 1: Is AI really creative, or is it just combining patterns it has seen before?

Question 2: If a human and an AI work together on a painting, who is the artist?

Question 3: Would you use AI to help with your own art projects? Why or why not?

Write your answer in 3–4 sentences and share with the class.

Quick Check ✓

1. In your own words, explain what AI is. How does it learn?
2. Name TWO AI tools that create visual art and describe what each one does.
3. What are TWO things humans do better than AI in creative work?
4. Describe the 'Next Rembrandt Project' and explain what it demonstrates about AI.
5. Do you think a painting made entirely by AI should be called 'art'? Explain your thinking.

Lesson Summary

- ★ AI is a computer program that learns from millions of examples and creates new content.
- ★ AI can generate art, music, stories, and videos from text prompts.
- ★ Humans bring feelings, experiences, and intention; AI brings speed and pattern-mixing.
- ★ Famous AI art tools include DALL·E, Midjourney, AIVA, and ChatGPT.
- ★ The future is humans and AI creating together — your imagination remains the most powerful tool!